# R Lab 6. Simultaneous Estimation

**(Sec. 4.1 and 30.2 [in Applied Linear Statistical Models book])**

```
> reg = lm( mpg ~ wt, data = mtcars )
> b = coef(reg)
> V = vcov(reg)
>
> # Joint 95% confidence ellipse
> library(ellipse)
> E = ellipse( V, centre = b, level = 0.95,
               t = sqrt(qf(0.95, 2, df.residual(reg)) * 2 ))
> plot(E, col="blue", lwd=3, type="l")
>
> # Add the 95% Bonferroni confidence rectangle
> CI = confint(reg, level = 0.975)
> rect( CI[1,1], CI[2,1], CI[1,2], CI[2,2], border="red", lwd=2 )
>
> # Add the vector of estimated coefficients
> points( b[1], b[2], lwd=5 )
```

